

Gergely Szabo

BACKEND & SYSTEMS ENGINEER

Gyor, Hungary

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Professional Summary

I am a Software Engineer mainly specialised in blockchain dApp and web backend development.

Work Experience

Haskell/Cardano Blockchain Consultant

remote

MLABS

Aug 2021 - present

- Led design and implementation of an off-chain tool in Haskell for building, batching and submitting transactions, handling millions of dollars worth of tokens safely.
- Developed a Rust-based off-chain toolkit including a transaction builder, chain-follower and indexer, improving development workflows and computational resource requirements. (Rust, PostgreSQL)
- Engineered Rust code generation for LambdaBuffers schema language compiler, facilitating type-safe integration in a polyglot environment.
- Worked as technical lead and project manager delivering Plutus smart contracts and CLI tools for IOG's partnerchain project, following agile methodologies with weekly sprint meetings and story points.
- Designed and implemented several Cardano dApps (Plutus, Plutarch, Servant, REST, gRPC)
- Co-designed and implemented an extension to the Agora on-chain voting platform, integrating with L1 dRep voting.
- Audited dApps finding security vulnerabilities and suggesting solutions.
- Implemented and scaled OpenFGA-based authorization across backend workflows.
- Refactored Haskell backend modules toward ports-and-adapters architecture.

Full Stack Web Engineer

remote

KAKEKOMU INC.

Jan 2018 - Jul 2021

- Designed and implemented a Q&A service backend/frontend using TypeScript, Koa.js, MySQL and Next.js, deployed to AWS ECS.
- Configured CI/CD flows for several services.
- Implemented the front end of a lawyer search platform in Next.js.
- Created an internal admin tool using Elm and Ruby on Rails.
- Built an internal UI library using React, published to a private GitHub npm registry.
- Worked in a remote team using agile methodologies.

Frontend Web Engineer

Tokyo, Japan

YAHOO JAPAN

Jan 2018 - Mar 2019

- Implemented new features on an advertisement management tool using React.
- Configured CI flow for the advertisement management tool using Screwdriver.
- Worked in both waterfall and agile workflows during the company's agile transition.

Junior Software Engineer

Tokyo, Japan

HAPPINESS TECHNOLOGY

Apr 2017 - Dec 2017

- Implemented new features on an inventory management system (Java, Oracle PL/SQL, Flash, JavaScript).

Projects

LambdaBuffers

Haskell / Rust / Nix

SCHEMA LANGUAGE AND CODE GENERATOR FOR POLYGLOT PROJECTS

<https://github.com/mlabs-haskell/lambda-buffers>

Plutus Ledger API

Rust

PLUTUS TYPES AND USEFUL TOOLS FOR CARDANO DAPP DEVELOPMENT

<https://github.com/mlabs-haskell/plutus-ledger-api-rust>

Tx Village	Rust
RUST BASED TOOLKIT FOR CARDANO TRANSACTION BUILDING, VERIFICATION AND CHAIN-INDEXING https://github.com/mlabs-haskell/tx-village	
Bot Plutus Interface	Haskell
TRANSACTION BUILDER FRAMEWORK BASED ON PLUTUS-APPS CONTRACT MONAD INTERFACE https://github.com/mlabs-haskell/bot-plutus-interface	
Plutip	Haskell
CARDANO LOCAL TEST NETWORK EXECUTER AND TEST FRAMEWORK https://github.com/mlabs-haskell/plutip	
Cardano Devnet Flake	Nix
CARDANO LOCAL TEST NETWORK EXECUTER BASED ON PROCESS-COMPOSE https://github.com/mlabs-haskell/cardano-devnet-flake	
Flake-lang	Nix
NIX TOOLS POWERING POLYGLOT MONO-REPOSITORIES https://github.com/mlabs-haskell/flake-lang.nix	
Szabo Gergely portfolio	Elm
PERSONAL PORTFOLIO WEB PAGE, SIMULATING A TERMINAL WITH A FEW COMMANDS LIKE `ECHO` AND `FIGLET`. https://github.com/szg251/szabo-gergely-portfolio	
Activity Analyser	Rust
FIT FILE ANALYSER FOR CYCLING ACTIVITIES https://github.com/szg251/activity-analyser	
Chess Clock	Rust (embedded)
CHESS CLOCK IMPLEMENTED FOR AN STM32 MICROCONTROLLER https://github.com/szg251/chesschock	

Education

Kyoto University of Education	Kyoto, Japan
JAPANESE LANGUAGE AND CULTURAL STUDIES EXCHANGE STUDENT 2013-2014	
A Tan Kapuja Buddhista College	Budapest, Hungary
BA IN JAPANESE LANGUAGE AND CULTURE 2011 - 2015	

Skills

Programming languages	I have several years of experience using Rust, Haskell, PureScript and TypeScript, but also worked on projects with Elm, Ruby and Python.
Backend development	Designed and built REST and gRPC backend services with the following libraries (e.g. Rocket, Axum, Tonic, Servant, IHP, Koa.js, Ruby on Rails.)
Databases	Mostly but not exclusively used relational databases like PostgreSQL for backend applications and blockchain indexers.
Blockchain dApp development	Designed and implemented several Cardano dApps, including sidechains, voting protocols, etc.
Frontend development	Worked as a full stack engineer maintaining large scale frontend and backend applications, using React (TypeScript) and Elm.
DevOps	Configured and maintained projects using Nix (Hercules CI), bare metal Linux, AWS (ECS, Aurora, Lambda, S3, etc.), GCP
Agentic Development	Used LLMs (e.g. Claude, Codex) to supplement my engineering workflows and to help my research on new topics, or create quick prototypes or PoCs.

Languages

Fluent	Hungarian (native), English (TOEIC 985), Japanese (JLPT N1)
Beginner	French, Italian

Interests

road cycling, chess, programming, bass guitar (jazz/fusion), language learning